

Fourth Semester, B.Tech Course work
Mid Semester Theory Examination, February, 2026

Course code: CBMTC13 & CIMTC13
Course title: Probability and Stochastic Processes

Maximum Marks: 25

Time: 1 hour 30 minutes

Note: Attempt all questions in given order only. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO										
1a	In a group of engineering students, 60% know MATLAB, 50% know Python, and 72% know at least one of these two software tools. What percentage of students know Python among those who also know MATLAB?	2.5	CO1										
1b	Show that the function $f_X(x) = \begin{cases} k(1 - x^2), & -1 \leq x \leq 1, \\ 0, & \text{otherwise,} \end{cases}$ is a probability density function. Find the value of k and obtain the distribution function of X .	2.5	CO1										
2a	A discrete random variable X takes values $\{-2, 0, 2\}$ such that $P(X = -2) = P(X = 2) = 2P(X = 0)$. Find the mean of the random variable $Y = 3X + 4$.	2.5	CO1										
2b	The probability density function of a random variable X is given by $f_X(x) = \begin{cases} \frac{1}{a}, & 0 \leq x \leq a, \\ 0, & \text{otherwise.} \end{cases}$ Find the variance of X .	2.5	CO1										
3a	The lifetime (in hours) of a machine component is a continuous random variable X with p.d.f. $f_X(x) = \begin{cases} \frac{200}{x^2}, & x \geq 200, \\ 0, & \text{otherwise.} \end{cases}$ Find the probability that the component lasts less than 400 hours, given that it has already lasted 300 hours.	2.5	CO1										
3b	A sample of 10 items is drawn at random from a box containing 60 items, of which 4 are defective. Find the probability that the sample contains at most one defective item.	2.5	CO2										
4a	Derive the mean of a Poisson distribution with parameter λ .	2.5	CO2										
4b	A random variable X is normally distributed with mean 50 and standard deviation 10. Find $P(40 \leq X \leq 70)$.	2.5	CO2										
5a	State and apply Chebyshev's inequality to find an upper bound for $P(X - 5 \geq 3)$, given that $E(X) = 5$ and $\text{Var}(X) = 4$.	2.5	CO2										
5b	Fit a Poisson distribution to the following data and obtain the expected frequencies. <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>x</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> </tr> <tr> <td>f</td> <td>90</td> <td>65</td> <td>30</td> <td>15</td> </tr> </table>	x	0	1	2	3	f	90	65	30	15	2.5	CO2
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